Akash Arvind Kore

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Work Authorization: F-1 OPT (STEM extension filed) | Open to Relocation

Summary

3+ years of experience building production AI systems and cloud-native infrastructure: **MCP servers**, **agentic workflows** (LangChain, LangGraph), **RAG pipelines**, and **LLM fine-tuning** (LoRA/QLoRA). Shipped systems delivering **90% reduction in ML review time**, **35% inference cost savings**, and **99.9% uptime**. Full stack across AI/ML engineering, backend systems, and cloud deployment on AWS. M.S. Computer Science, NC State University.

Experience

AI Software Engineer Aug. 2025 – Present

Ready Tensor, Inc. *San Diego, CA (Remote)*

- Built and shipped **readytensor** (PyPI): an **LLM evaluation and observability framework** scoring AI/ML repos across 85 criteria via GPT-4o with **structured outputs** (Instructor) and seeded, deterministic inference – cutting ML review time by **90%**.
- Engineered a **multi-LLM routing layer** (OpenAI, Gemini, Groq, Anthropic, OpenRouter) with **function calling**, **tool use**, and **LangChain** orchestration; reduced inference costs by **35%** across a 10-person team.
- Built production **MLOps pipelines** (Docker, AWS Lambda, GitHub Actions CI/CD) achieving **99.9% uptime**; executed zero-downtime migrations and established IaC standards adopted team-wide.
- Developed and shipped a full **authentication system** (OAuth, email verification, RBAC, localization) supporting a platform-wide migration for **50,000+ users**.

Machine Learning Intern May 2024 – Aug. 2024

Japan Atomic Energy Agency (JAEA) – Japan's national nuclear research lab *Remote*

- Designed a **computer vision** pipeline for nuclear reactor bubble detection; selected **Pix2Pix GAN** (PyTorch) for sim-to-real **domain adaptation** under limited labeled-data constraints – **90% detection accuracy**, **30% improved generalization**.
- Curated a 50k-image labeled dataset; applied **transfer learning** fine-tuning; containerized the **real-time inference pipeline** (Docker, AWS Lambda) achieving **15 FPS** and **95% overhead reduction**.

Software Engineer Associate Sep. 2021 – Jun. 2023

Siemens Digital Industries Software *Pune, India*

- Engineered Teamcenter on Cloud (TcX) on **AWS (EC2, S3, RDS, Lambda)** for **500+ global enterprise customers**; automated CI/CD via CloudFormation, cutting deployment setup time by **30%**.
- Resolved **40+ critical production incidents** across Python, C++, and Bash; refactored **C++ server-side** components – reducing bug resolution time by **25%**.

Projects

B.I.A.S.E.D. – Agentic RAG Decision Intelligence Platform github.com/TheSelfSa/biased *Nov. 2024 – Mar. 2025*

FastAPI, Next.js, pgvector, LangGraph, LangChain, MCP, Ollama, Docker | Apache-2.0

- Built an **MCP server** enabling external tool integration; implemented **function calling** and **tool use** within a **LangGraph** multi-agent orchestration layer – routing queries across LLMs and returning **structured outputs** via **pgvector** RAG.
- Full-stack open-source platform: Next.js PWA + FastAPI backend, **RBAC** auth, Docker Compose, scenario planning engine for multi-hypothesis decision support; Apache-2.0.

End-to-End LLM Fine-Tuning Pipeline *Jun. 2024 – Sep. 2024*

PyTorch, Hugging Face Transformers, LoRA/QLoRA (PEFT), vLLM, Weights & Biases, Ray Tune

- Fine-tuned **LLaMA-3** with **LoRA/QLoRA (PEFT)** on domain-specific data; optimized hyperparameters via **Ray Tune** and tracked with **Weights & Biases**; evaluated against baseline using ROUGE-L and BERTScore.
- Deployed quantized **inference endpoints** via **vLLM** + FastAPI achieving **40% memory footprint reduction**; production template for rapid **foundation model** adaptation and **inference optimization**.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Bash

Agentic & Tooling: MCP (Model Context Protocol), LangChain, LangGraph, Function Calling, Tool Use, Agentic Workflows, Multi-Agent Systems, Claude Code

LLMs & GenAI: GPT-4o, LLaMA-3, Claude (Anthropic), Gemini, OpenAI API, LLM Fine-Tuning (LoRA/QLoRA), PEFT, Retrieval-Augmented Generation (RAG), Model Evaluation, LLM Observability, vLLM, Inference Optimization

AI / ML: PyTorch, TensorFlow, Hugging Face Transformers, Deep Learning, NLP, Computer Vision, Scikit-Learn, Distributed Training, ONNX, TensorRT, Ray

MLOps & Cloud: AWS (EC2, S3, RDS, Lambda), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, SageMaker, MLflow, Weights & Biases, IaC

Data & Vectors: Vector Databases, pgvector, Pinecone, ChromaDB, FAISS, PostgreSQL, MongoDB, Redis, Kafka

Web & APIs: FastAPI, Django, Next.js, React.js, REST APIs, GraphQL, Pydantic, TailwindCSS

Education

Master of Science, Computer Science Aug. 2023 – May 2025

North Carolina State University *Raleigh, NC*

Bachelor of Technology, Computer Science Aug. 2017 – Jul. 2021

Walchand College of Engineering *Sangli, MH, India*